

USHARANI VARAKAPADRA

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Career Objective

Computer Science undergraduate with a strong foundation in Python and hands-on experience building AI/ML and LLM-based applications. Comfortable working with data using Pandas, NumPy, and SQL, with a growing interest in NLP and applied machine learning through project-based learning.

Education

B.Tech – Computer Science and Engineering	Oct 2023 – Present
<i>Rajiv Gandhi University of Knowledge Technologies, Nuzvid</i>	<i>CGPA: 8.34</i>
Pre University Course (PUC)	Nov 2021 – July 2023
<i>Rajiv Gandhi University of Knowledge Technologies, Nuzvid</i>	<i>CGPA: 9.75</i>
Secondary School Certificate	2020 – 2021
<i>Sri Gurukula Vidyalayam</i>	<i>CGPA: 10.0</i>

TECHNICAL SKILLS

Languages: Python, C, SQL

AI/ML: Machine Learning, Artificial Intelligence, NLP, LLMs (basic), Deep Learning (Basic)

Data: Pandas, NumPy, Data Cleaning, Data Visualization

Web: HTML, CSS, JavaScript

Tools: GitHub, Jupyter Notebook, Visual Studio Code

Projects

AI-Based Student Analysis Agent

- Developed an LLM-powered educational assistant using LLaMA 3 via Groq API to generate curriculum-aware MCQs, structured explanations, and personalized recommendations through prompt engineering.
- Engineered structured prompts for deterministic MCQ generation and consistent JSON-formatted responses across Data Structures, Algorithms, DBMS, OS, CN, and OOP.
- Built an analytics dashboard tracking strong/weak topics and score history, visualized with Chart.js.
- Designed a multi-step prompting workflow that analyzes performance history to generate personalized study recommendations.

Technologies: React.js, Groq API, LLaMA 3, Prompt Engineering, JavaScript, localStorage, Chart.js

AI-Powered Project Idea Generator & Mentor

- Built a full-stack Flask web application that generates personalized, LLM-driven project recommendations tailored to a user's skills, experience level, time constraints, and career goals, addressing the gap between learning concepts and applying them practically
- Designed a scoped AI mentorship system that provides step-by-step guidance, code review, and contextual hints for each project milestone, using conversation history to maintain relevant context per task
- Implemented a dataset resolution pipeline that verifies and returns real, working dataset links (Kaggle/data.gov) instead of relying on LLM-generated URLs, preventing broken/hallucinated references
- Structured relational data models (Project, Milestone, Mentor Chat) using Flask-SQLAlchemy to track user progress across generated projects and milestones

Technologies: Python, Flask, Flask-SQLAlchemy, SQLite, Groq API (Llama 3.3 70B), Jinja2, Bootstrap 5

Detecting Anomalies in ATM Transactions using QML

- Built an anomaly detection model using Quantum Machine Learning (QML) to flag suspicious ATM transactions.
- Implemented quantum circuits with Qiskit and Python to analyze transaction patterns.
- Performed data preprocessing, feature extraction, and model evaluation.

Technologies: Python, Quantum Machine Learning (QML), Qiskit, NumPy, Scikit-learn

Certifications

- Python
- Foundation Level in Programming and Data Science, IITM
- Foundations of Deep Learning – NPTEL
- Cloud Computing-Nptel